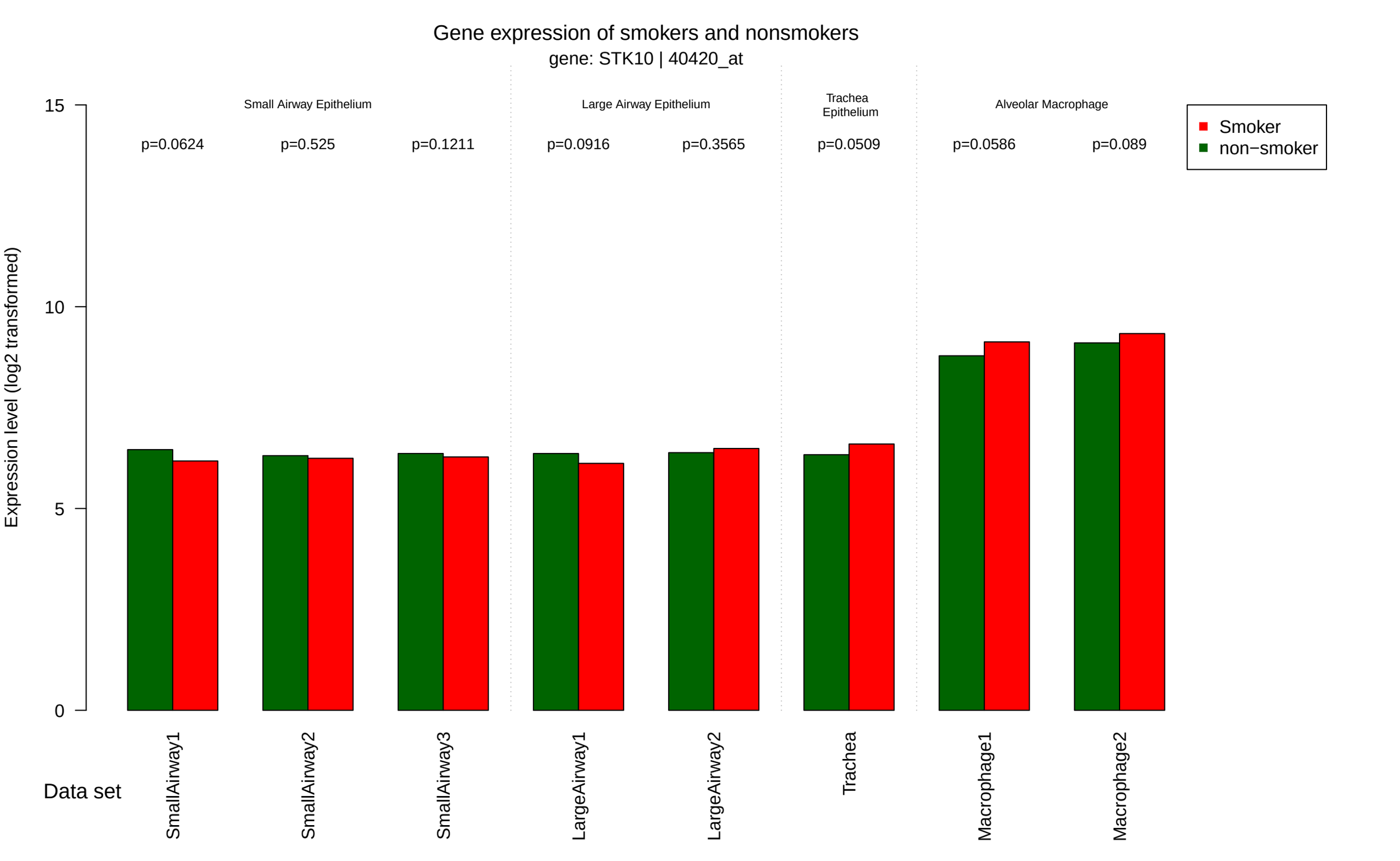
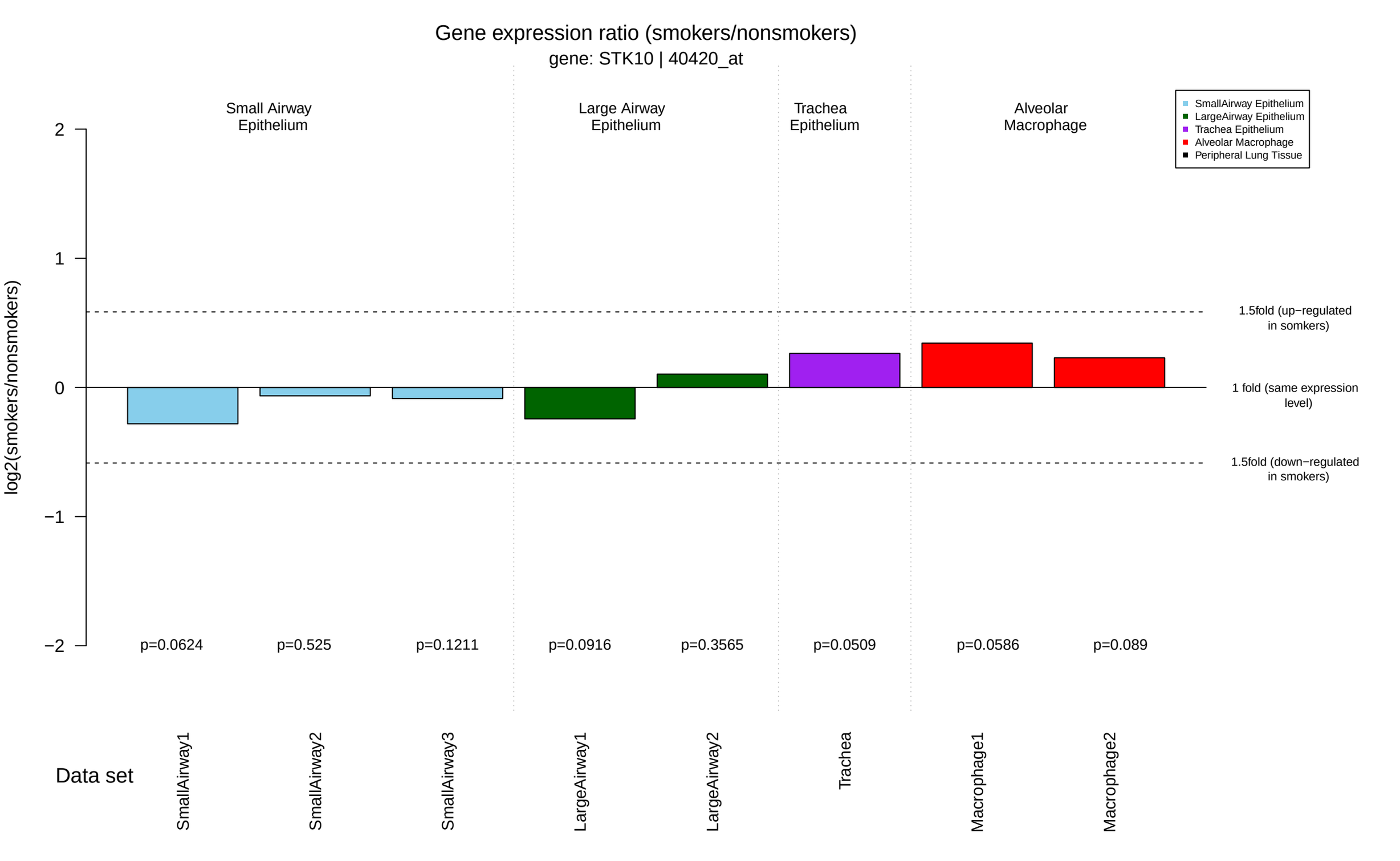
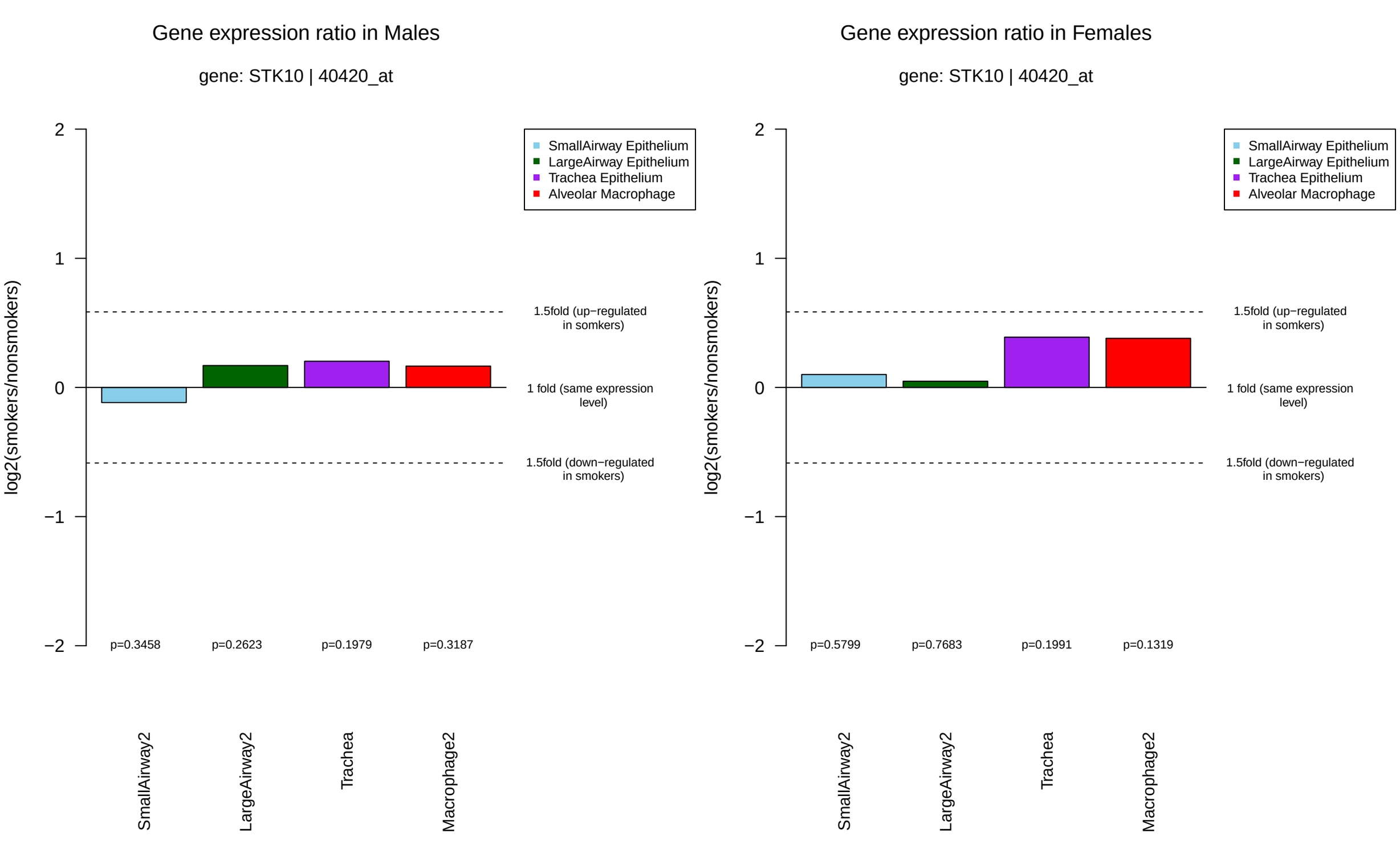
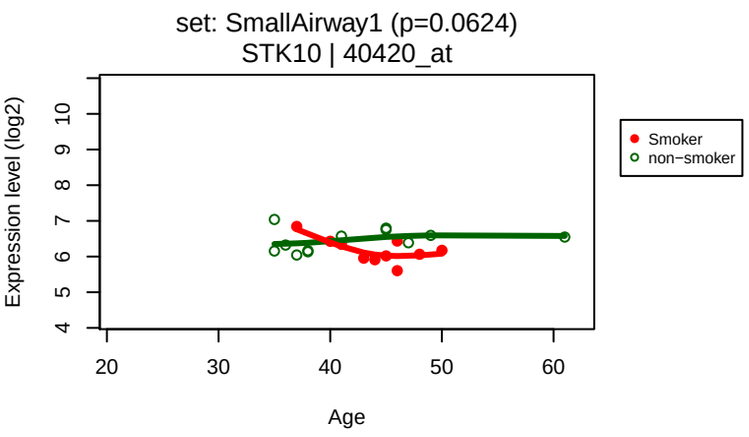
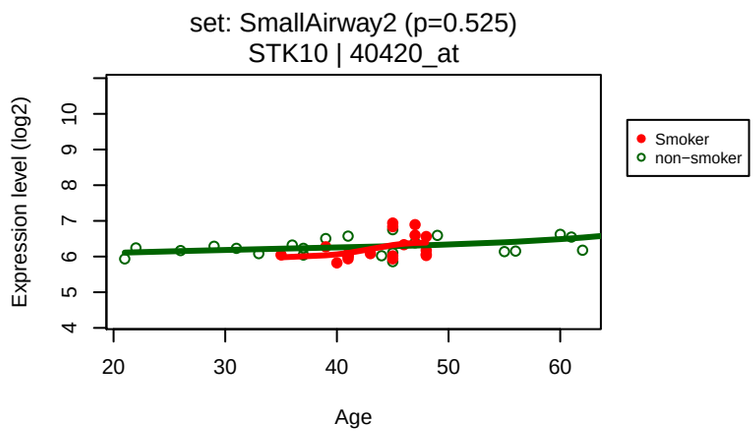
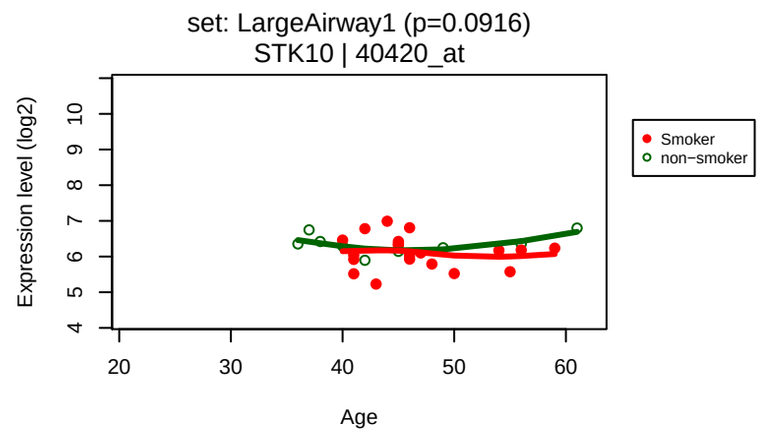

Gene expression of smokers and nonsmokers
gene: STK10 | 40420_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0624
p=0.525
p=0.1211
p=0.0916
p=0.3565
p=0.0509
p=0.0586
p=0.089
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

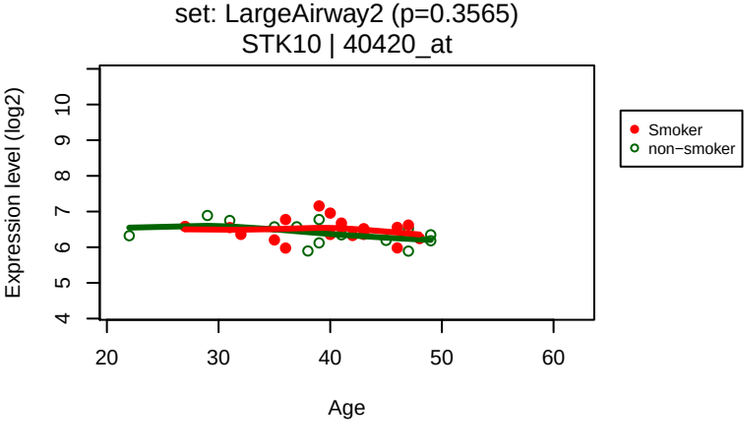

Gene expression ratio (smokers/nonsmokers)
gene: STK10 | 40420_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0624
p=0.525
p=0.1211
p=0.0916
p=0.3565
p=0.0509
p=0.0586
p=0.089
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

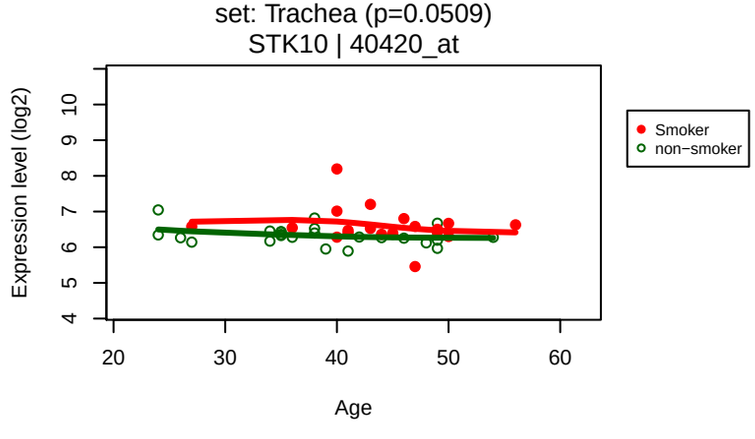

Gene expression ratio in Males
gene: STK10 | 40420_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3458
p=0.2623
p=0.1979
p=0.3187
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STK10 | 40420_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5799
p=0.7683
p=0.1991
p=0.1319
SmallAirway2
LargeAirway2
Trachea
Macrophage2

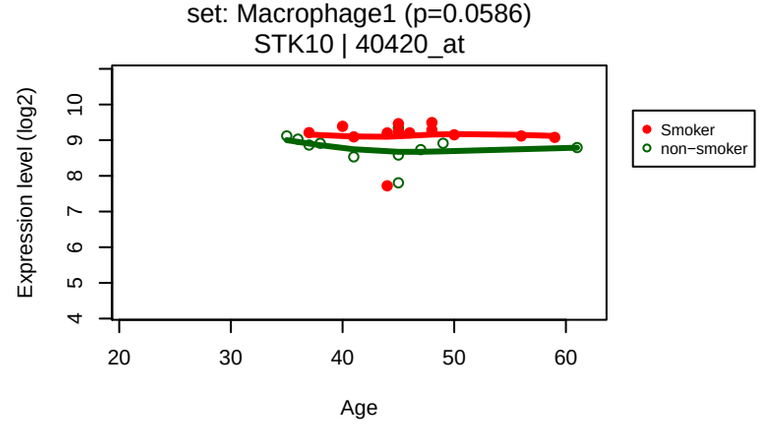

set: SmallAirway1 (p=0.0624)
STK10 | 40420_at
Expression level (log2)
Age
Smoker
non−smoker

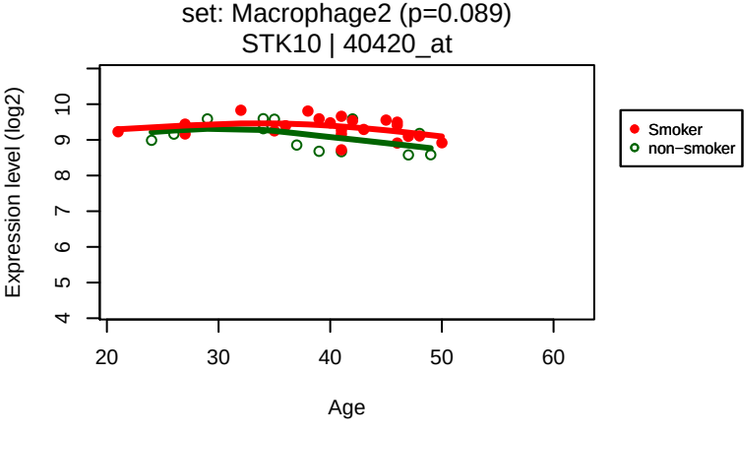

set: SmallAirway2 (p=0.525)
STK10 | 40420_at
Expression level (log2)
Age
Smoker
non−smoker

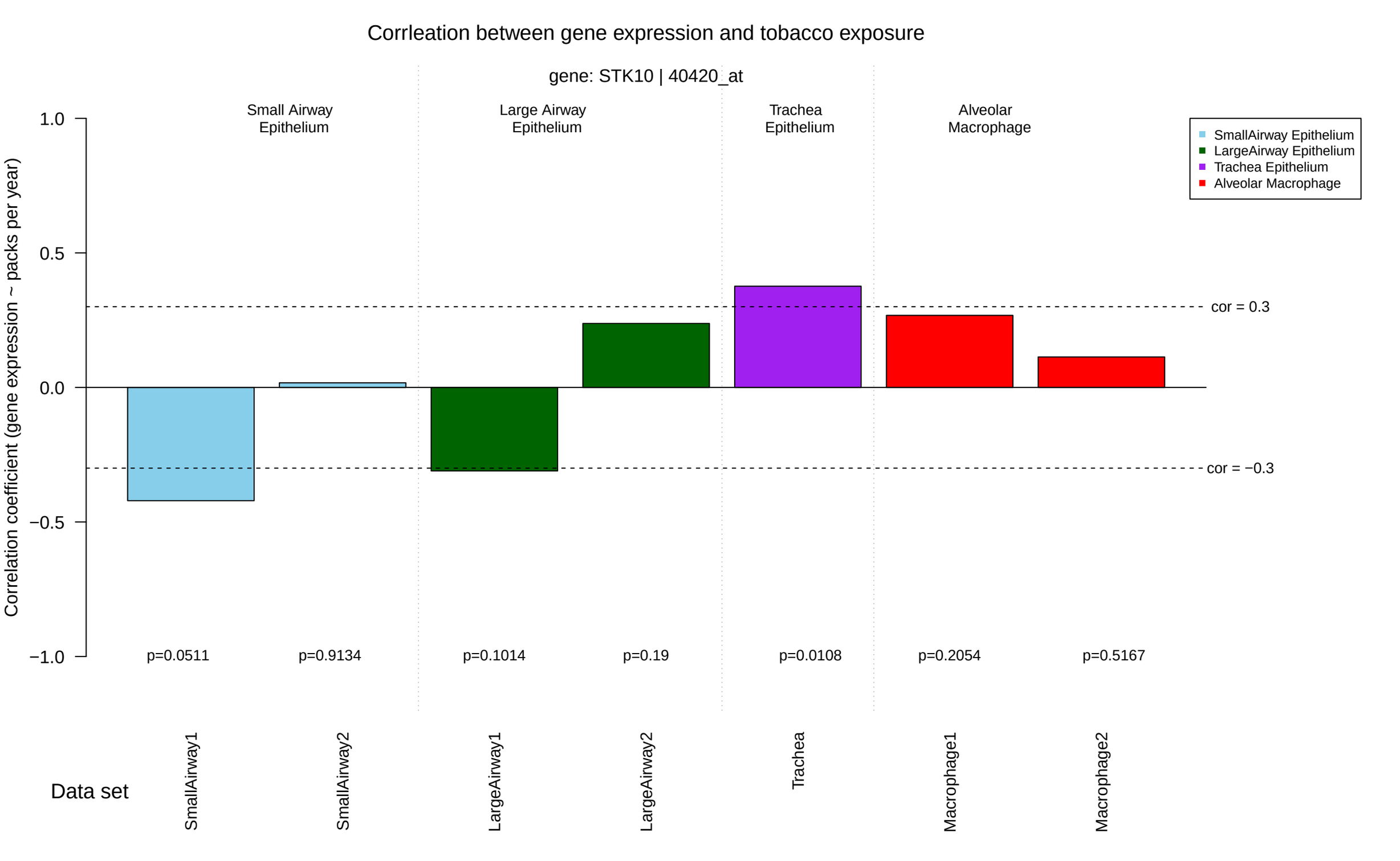

set: LargeAirway1 (p=0.0916)
STK10 | 40420_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3565)
STK10 | 40420_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0509)
STK10 | 40420_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0586)
STK10 | 40420_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.089)
STK10 | 40420_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STK10 | 40420_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0511
p=0.9134
p=0.1014
p=0.19
p=0.0108
p=0.2054
p=0.5167
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2